

```
align(center)[ text(fill: black90, size: 11pt, weight: "bold")[title] linebreak() text(fill: black70, size: 8pt)
[subtitle] linebreak() text(fill: black70, size: 6.8pt)[Five DRIVE training images. Filtering uses the
green channel only. Noisy metrics average over 10 AWGN draws.] ] v(8pt) image-grid(data,
[Gradient magnitude, clean retinal images], "magnitude_path", show-input: true)
```

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align(center)[ text(fill: black90, size: 11pt, weight: "bold")[title] linebreak() text(fill: black70, size: 8pt)
[Orientation maps. Hue encodes edge angle modulo  $\pi$  and brightness follows normalized
magnitude.] ] v(8pt) image-grid(data, [Gradient orientation, clean retinal images],
"orientation_path", show-input: false)
```

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align(center)[ text(fill: black90, size: 11pt, weight: "bold")[title] linebreak() text(fill: black70, size: 8pt)
[Quantitative summary on vessel-boundary ODS, boundary-vector RMSE, and centerline tangent
orientation MAE.] ] v(8pt) grid( columns: 2, column-gutter: 10pt, [ metric-chart(data, "ods_f_score",
[Soft-label ODS F-score versus SNR], [ODS F-score]) ], [ metric-chart(data,
"gradient_vector_rmse_mean", [Gradient-vector RMSE versus SNR], [Vector RMSE]) ], [ metric-
chart(data, "orientation_mae_deg_mean", [Centerline tangent MAE versus SNR], [Angle MAE
(deg)]) ], [ summary-table(data) ], ) v(10pt) legend(data)

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